A Bayesian Boosted Approach to Predicting NCAA Tournament Outcomes

Using Seed Priors, XGBoost Regression, and Round-Specific Probability Calibration

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Introduction

March Madness is one of the most popular post-season tournaments in the country because of its unique format. Every year, 64 of the best college basketball teams compete in a single-elimination tournament, where one bad game can end a team's season. Unlike many other sports, this format makes March Madness extremely difficult to predict because there is so much variance in the results. In 2025, 26.6 million brackets were filled out on ESPN alone, but not a single one was perfect. In fact, a perfect bracket has never been recorded in the history of March Madness. With the help of machine learning, we can make more educated predictions and better understand what separates predictable games from games where upsets are more likely.

One major challenge in predicting March Madness outcomes is that the landscape of college basketball is constantly changing. In recent years, NIL and the transfer portal have changed how teams build their rosters. Players now have more freedom to transfer, and schools with stronger NIL resources may have a better chance of attracting and retaining high-level talent. This creates a new challenge for prediction models because historical tournament data may not fully reflect the current era of college basketball.

This shift was especially clear in the 2025 tournament. There was a record-setting lack of upsets, and for the first time in tournament history, all four number one seeds reached the Final Four. With this in mind, the goal of this project was to create a model that can accurately predict historical March Madness games, while also being flexible enough to predict future tournaments in the changing NIL and transfer portal era.

The final model predicts the expected point differential for each game. From there, the predicted margin is converted into a win probability using a normal distribution. Instead of using the same standard deviation for every game, the model uses round-specific sigma values because early-round games tend to have more variance and larger score differences than later-round games.

Research Questions

- Can a Bayesian-boosted machine learning model accurately predict the winner of NCAA Tournament games?
- Can the same model estimate the expected margin of victory for each matchup?
- Does the model outperform a seed-prior baseline and sportsbook opening spreads?

Data

There were two main sources of data used in this model. The primary source was Ken Pomeroy's college basketball analytics site, which provided season-long adjusted efficiency data for every tournament team. The model uses KenPom data starting in 2003, so the final dataset includes NCAA Tournament games from 2003 through 2026.

From KenPom, three main efficiency variables were collected: adjusted offensive efficiency, adjusted defensive efficiency, and adjusted efficiency margin. These numbers are adjusted for strength of schedule, which makes them especially useful for March Madness because teams come from very different conferences.

The second data source was Kaggle's March Mania Machine Learning Competition. This dataset provided the results of every NCAA Tournament game, along with the seeds of the teams in each matchup. The model uses seed as a Bayesian prior: the historical win probability of each seed matchup gives the model a baseline expectation, and the XGBoost model then adjusts that expectation based on the actual quality of the teams.

The final model was trained as a margin prediction model. The six variables used in the final feature set are described in Table 1 in the appendix.

The last variable, "seed surprise difference," is especially important because it helps the model identify teams that may be overseeded or underseeded. It compares each team's

adjusted efficiency margin to the historical average for teams with the same seed. The seed surprise difference is calculated as:

```
diff_surprise = (AdjEM_W − E[AdjEM | seed_W]) − (AdjEM_L −
                    E[AdjEM | seed_L])
```

This allows the model to keep the seed prior as a baseline while still adjusting for teams whose actual efficiency profile does not match their seed.

Modeling

There are two main outputs from this project: predicted margin and win probability. Instead of building these as two separate models, the predicted margin model was created first, and the win probability was derived directly from it. This ensures both outputs are always consistent with each other.

The model was trained on historical NCAA Tournament games before 2025. The 2025 season was used as the main holdout test season, while 2026 was used as a future validation season. The final model was built in two steps: a Bayesian seed prior and a gradient-boosted regression model.

Bayesian Seed Prior

Figure1 in the appendix shows the Bayesian seed prior, used to estimate the probability that the better seed wins a given matchup. Each cell represents a historical seed pairing, with blue showing matchups where the better seed has historically won more often and red showing matchups where the worse seed has won more often.

The probabilities were calculated using a Laplace-smoothed estimate:

```
p = (wins + 1) / (total games + 2)
```

This smoothing prevents the model from overreacting to small sample sizes. After calculating the smoothed win probability, the prior was converted into log-odds and used as a feature in the margin prediction model.

Gradient Boosting Model

After creating the Bayesian seed prior, an XGBoost gradient-boosted regression model was used to predict the expected scoring margin for each tournament game. The model predicts margin rather than win/loss because margin contains more information. A 20-point win and a 1-point win both count the same in a classification model, but they tell very different stories about team strength.

XGBoost was chosen because it can capture nonlinear relationships and interactions between variables. Gradient boosting builds many small decision trees, where each tree tries to correct the mistakes of the previous trees. The model used a low number of trees (100) and a small learning rate (0.08) to avoid overfitting on the relatively small 63-game-per-year dataset.

Figure 3 in the appendix shows the feature importance from the final model. The most important feature was the seed prior log-odds at 36.1% of total split usage, followed by seed surprise difference at 18.9%. Team 2 adjusted efficiency margin contributed 15.6%, Team 1 adjusted defensive efficiency 13.3%, Team 1 adjusted efficiency margin 8.5%, and Team 1 adjusted offensive efficiency 7.6%.

Sigma Calibration

After the XGBoost regression model produced a predicted scoring margin, that margin was converted into a win probability using a normal distribution:

```
P(Team 1 wins) = Φ( Predicted Margin / σ )
```

A larger sigma means there is more assumed variance, so the same predicted margin produces a less confident win probability. A smaller sigma means less variance, so the same predicted margin produces a stronger win probability.

Sigma was calibrated separately for each round group because early-round games have more variance than late-round games. The calibration process tested a grid of sigma values from 5 to 25 in increments of 0.5, selecting the value with the lowest training log-loss. The Final Four and Championship were grouped together due to small sample sizes. The full sigma lookup is shown in Table 2 in the appendix and Figure 4.

Results

The results show that the model performed well as both a winner prediction model and a margin prediction model. The following table summarises the key metrics across all evaluation splits:

Metric	Result
Training Accuracy	**76.0%**
2025 Testing Accuracy	**84.1%**
2026 Validation Accuracy	**90.5%**
Training Brier Score	**0.15**
2025 Brier Score	**0.11**
2026 Brier Score	**0.11**
Training MAE	**8.0 pts**
2025 Testing MAE	**8.7 pts**
2026 Model vs. Sportsbook MAE	**9.58 vs. 9.70 pts**

Overall Performance

As shown in Figures 5 and 10 in the appendix, the model achieved an accuracy of 84.1% in 2025 and 90.5% in 2026. The training accuracy was 76%, above the historical

seed-only baseline of roughly 70%. Even in the most difficult training season, the model achieved 71.4%, consistently adding value beyond simply picking the better seed.

The higher testing and validation accuracy should be interpreted in context. The 2025 and 2026 tournaments had fewer upsets than many historical seasons. Because of this, accuracy in the mid-80% range is a more realistic long-run expectation, particularly given the changing NIL and transfer portal landscape.

Sportsbook Comparison

As shown in Dashboard 1 in the appendix, the model outperformed the sportsbook benchmark in winner prediction. The model had a 90.5% accuracy compared to 76.2% for the sportsbook. This suggests the model was not simply benefiting from a low-upset environment, but was making genuinely stronger winner predictions than the benchmark.

Probability Calibration

As shown in Figures 7 and 11, the Brier scores for 2025 and 2026 were both approximately 0.11, lower than the training Brier score of 0.15. This indicates that the model's probability predictions were better calibrated on holdout data than on training data. The log-loss results in Figures 6 and 11 support this further, suggesting the round-specific sigma calibration was effective.

The model's ability to identify close games was a particular strength. The Kentucky vs. Santa Clara game was predicted with a margin of 0.0, meaning the model viewed it as essentially a coin flip. The actual game went to overtime after a buzzer-beater finish.

Margin Prediction

As shown in Figures 9 and 12, the training MAE was 8.0 points and the testing MAE was 8.7 points. In the sportsbook comparison, the model's MAE was 9.58 points versus the sportsbook's 9.70 points; a small but meaningful margin in favour of the model.

Figure 13 shows the predicted vs. actual margin scatter plot. Most points fall on the positive side of both axes, meaning the model usually predicted the correct winner. The largest errors came from games with very large actual margins, which is expected because final margins can expand late due to fouling, bench scoring, and pace changes.

2026 Bracket Highlights

The model predicted Michigan to win the national championship. In the championship game, the model predicted Michigan to win by 6.3 points, and Michigan won by 6 points. The model also correctly identified several meaningful upsets: 11-seed Texas and 11-seed VCU in the first round, 6-seed Tennessee over 3-seed Virginia in the second round, and 9-seed Iowa over 4-seed Nebraska in the Sweet 16.

Conclusion

Overall, the results answer all three research questions. The Bayesian-boosted model accurately predicted NCAA Tournament winners in both 2025 and 2026. It estimated expected margins competitively with sportsbook benchmarks, slightly outperforming them in MAE. Most importantly, it outperformed the seed-prior baseline and showed significantly stronger winner prediction than the sportsbook comparison group.

The strongest result of the project is that combining historical seed matchup information with XGBoost efficiency adjustments and round-specific sigma calibration produced a model that was accurate, competitive, and interpretable.

Future Work

One major area for future work is adjusting the model to the new NCAA Tournament format. Reports indicate that the tournament is expected to expand from 68 to 76 teams starting in 2027. Under the proposed format, more bubble teams would enter the field, and 24 teams would play in opening-round games before the traditional first round begins. This would require the model to account for additional matchup types and lower-seeded teams that historically would have missed the tournament.

The most significant modeling impact is that the meaning of each seed line may change. Mid-major automatic qualifiers who previously landed around the 12-seed line could be pushed to the 13-seed line, while power-conference bubble teams may populate the 11–12 seed lines. The model was built to be adaptable because it does not

rely solely on seed, but starts with seed history and then adjusts using KenPom efficiency metrics and seed surprise. This should help the model respond to the new format more effectively than a pure seed-based model.

Future versions could also include separate sigma treatment for the new opening-round games, which may have different variance characteristics than traditional Round of 64 games. As more data becomes available, the model could estimate separate sigma values for those games.

Appendix

Tables

Table 1. Variable Description Table

Variable	Description
Prior Log Odds	The Bayesian seed prior converted into log-odds. Represents the historical win probability of the seed matchup before adding team-quality information.
Team 1 Adj. Offensive Efficiency	Team 1's KenPom adjusted offensive efficiency: how efficiently Team 1 scores per 100 possessions after adjusting for opponent strength.
Team 1 Adj. Defensive Efficiency	Team 1's KenPom adjusted defensive efficiency: how efficiently Team 1 prevents scoring per 100 possessions after adjusting for opponent strength.
Team 1 Adj. Efficiency Margin	Team 1's overall strength rating (AdjOE − AdjDE), adjusted for schedule.
Team 2 Adj. Efficiency Margin	Team 2's overall strength rating (AdjOE − AdjDE), adjusted for schedule.
Seed Surprise Difference	How much stronger or weaker each team is compared to the historical average team with the same seed. Captures over- and under-seeded teams.

Table 2. Round-Specific Sigma Values

Round Group	N (training games)	Best σ
Play-In	≈ 40	10.0
Round of 64 (R64)	≈ 660	10.0
Round of 32 (R32)	≈ 343	7.0
Sweet 16 (S16)	≈ 160	8.0
Elite Eight (E8)	≈ 88	6.0
Final Four + Champ.	≈ 63	5.0

Table 3. Accuracy and MAE by Season (2003-2026)

Season	N	Accuracy	Brier	LogLoss	MAE Points	Split
2003	63	73.0%	0.1493	0.4576	7.90	Training
2004	63	76.2%	0.1363	0.4181	7.00	Training
2005	63	77.8%	0.1384	0.4241	7.10	Training
2006	63	73.0%	0.1628	0.4826	7.68	Training
2007	63	81.0%	0.1207	0.3845	7.63	Training
2008	63	82.5%	0.1248	0.3870	9.21	Training
2009	63	74.6%	0.1383	0.4170	7.96	Training
2010	63	73.0%	0.1469	0.4438	6.67	Training
2011	63	73.0%	0.1828	0.5598	7.60	Training
2012	63	73.0%	0.1753	0.5351	7.44	Training
2013	63	73.0%	0.1549	0.4635	9.98	Training
2014	63	73.0%	0.1671	0.4999	7.90	Training
2015	63	79.4%	0.1311	0.4205	7.15	Training
2016	63	77.8%	0.1400	0.4269	7.97	Training
2017	63	81.0%	0.1238	0.3827	7.13	Training
2018	63	76.2%	0.1614	0.5199	7.94	Training
2019	63	74.6%	0.1325	0.4001	8.90	Training
2021	63	75.9%	0.1537	0.4639	7.21	Training
2022	63	71.4%	0.1874	0.5746	9.70	Training
2023	63	73.0%	0.1719	0.5575	8.39	Training
2024	63	76.2%	0.1653	0.4930	9.34	Training
2025	63	84.1%	0.1107	0.3490	8.70	Holdout
2026	63	90.5%	0.1163	0.4069	9.58	Holdout

Figures

Figure 1. Bayesian Seed Prior Heat Map

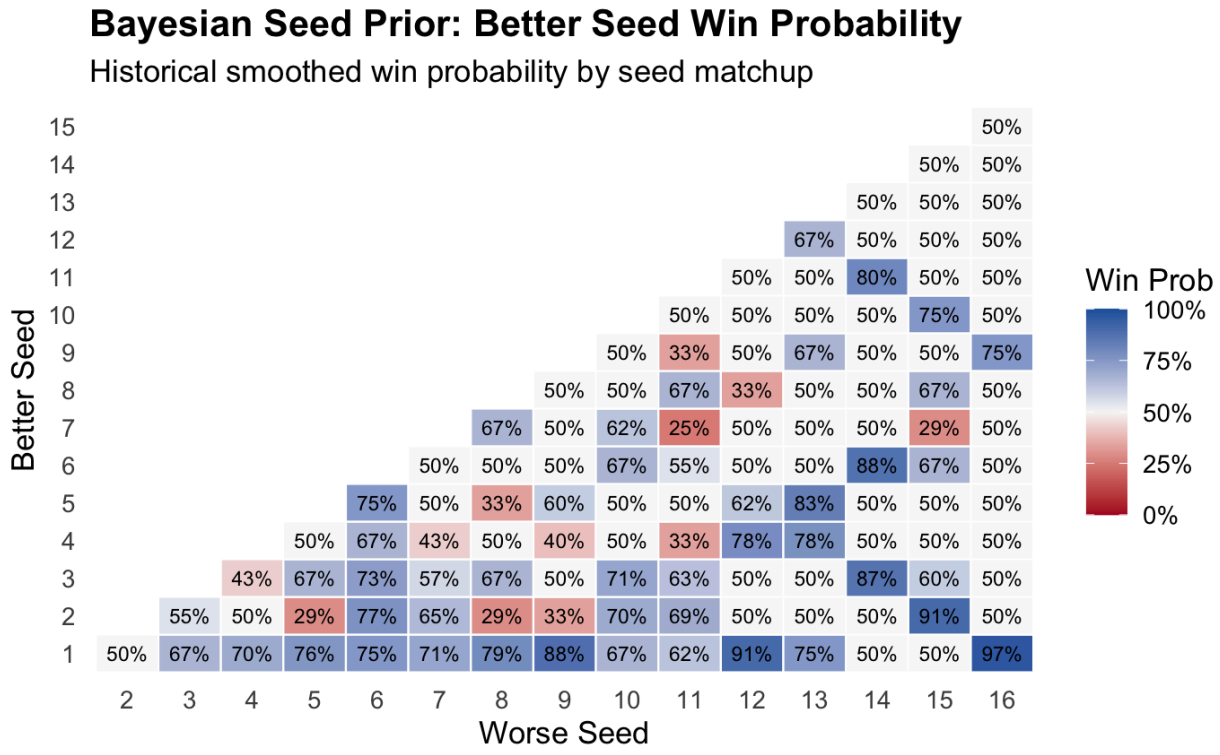


Figure 2. Baseline Prediction Margin by Seed Matchup

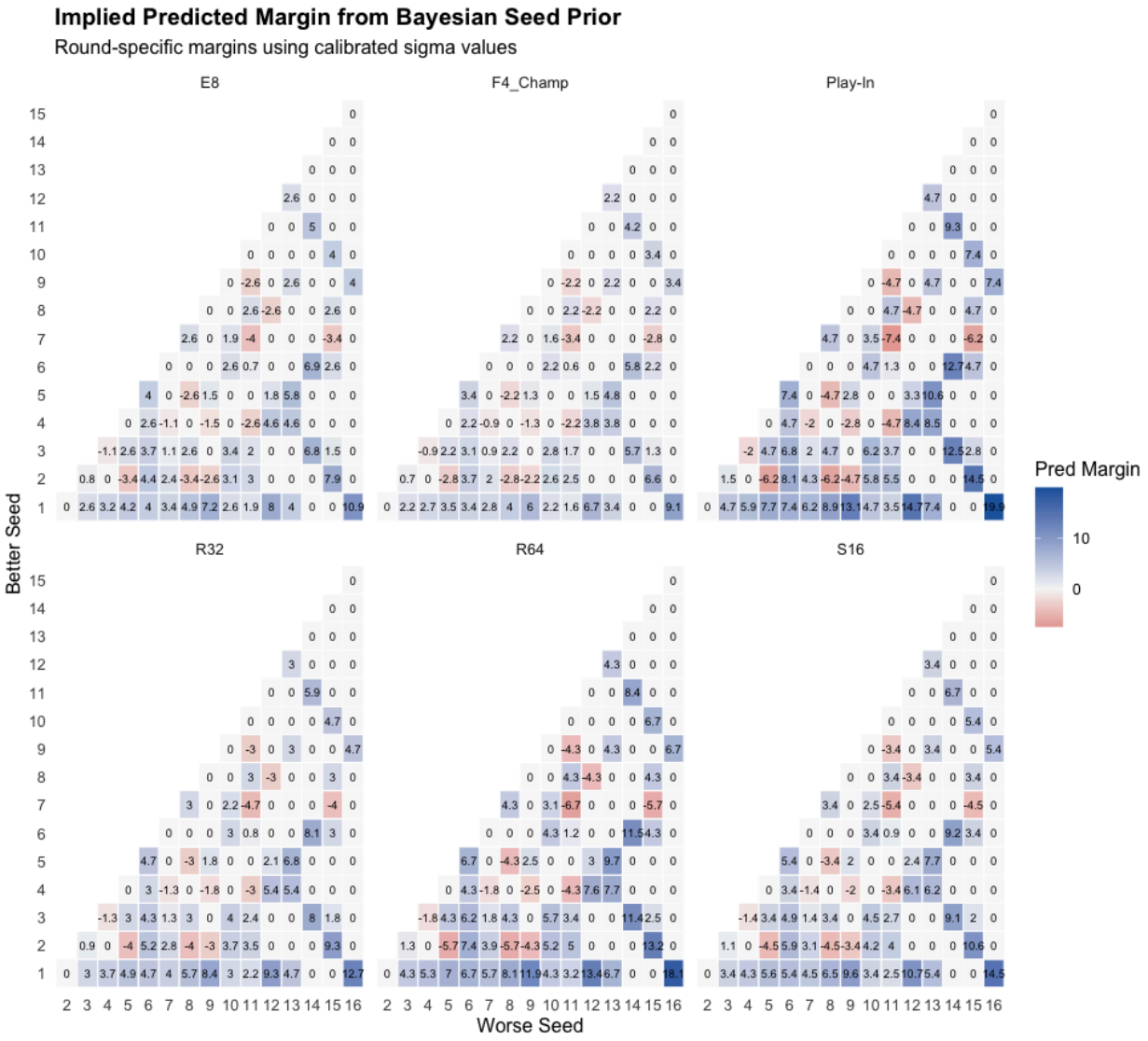


Figure 3. Feature Importance for XGBoost Regression Model

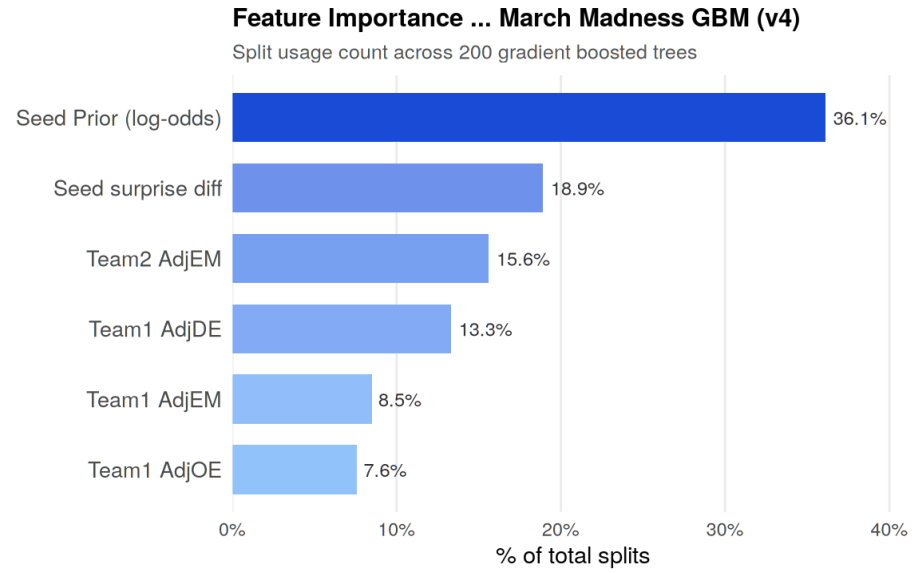


Figure 4. Sigma Calibration by Round

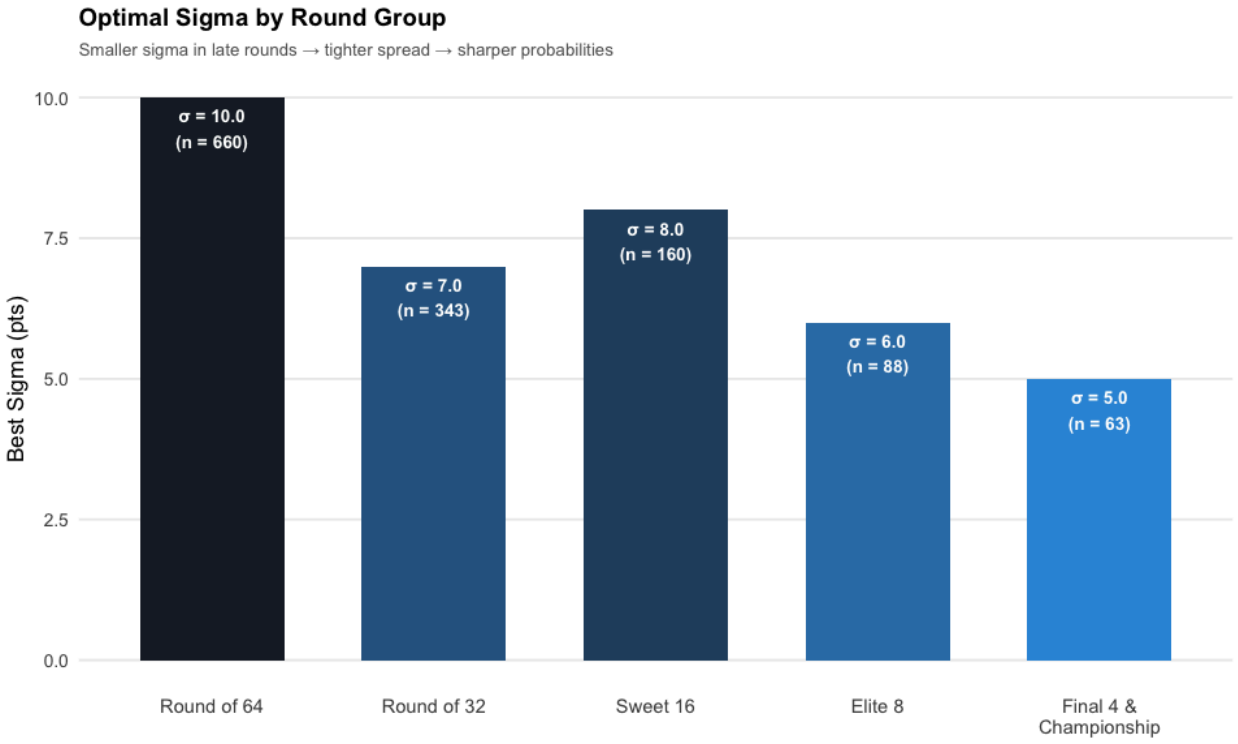


Figure 5. Accuracy by Round (2025 & 2026)

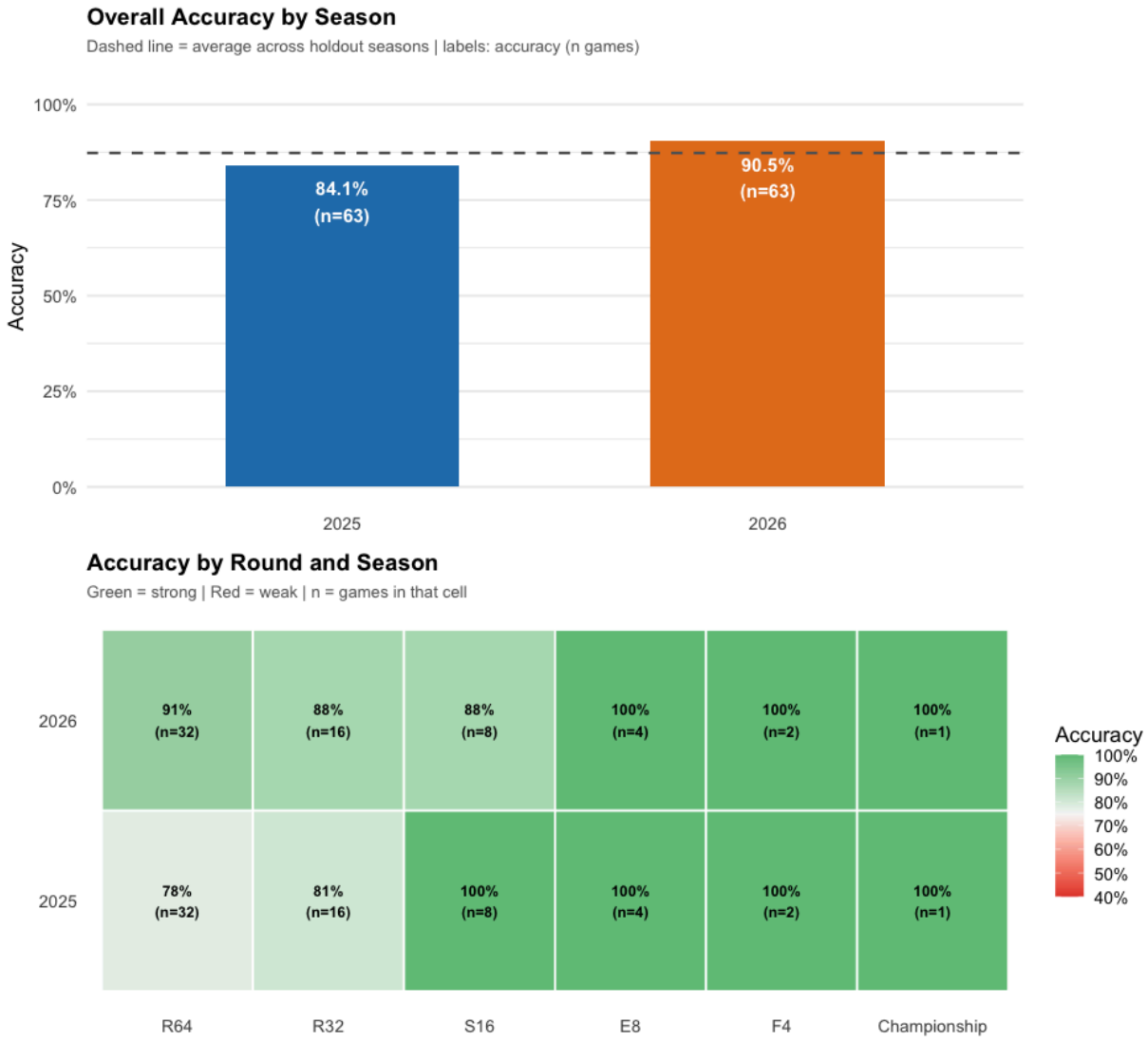


Figure 6. Log Loss by Round (2025 & 2026)

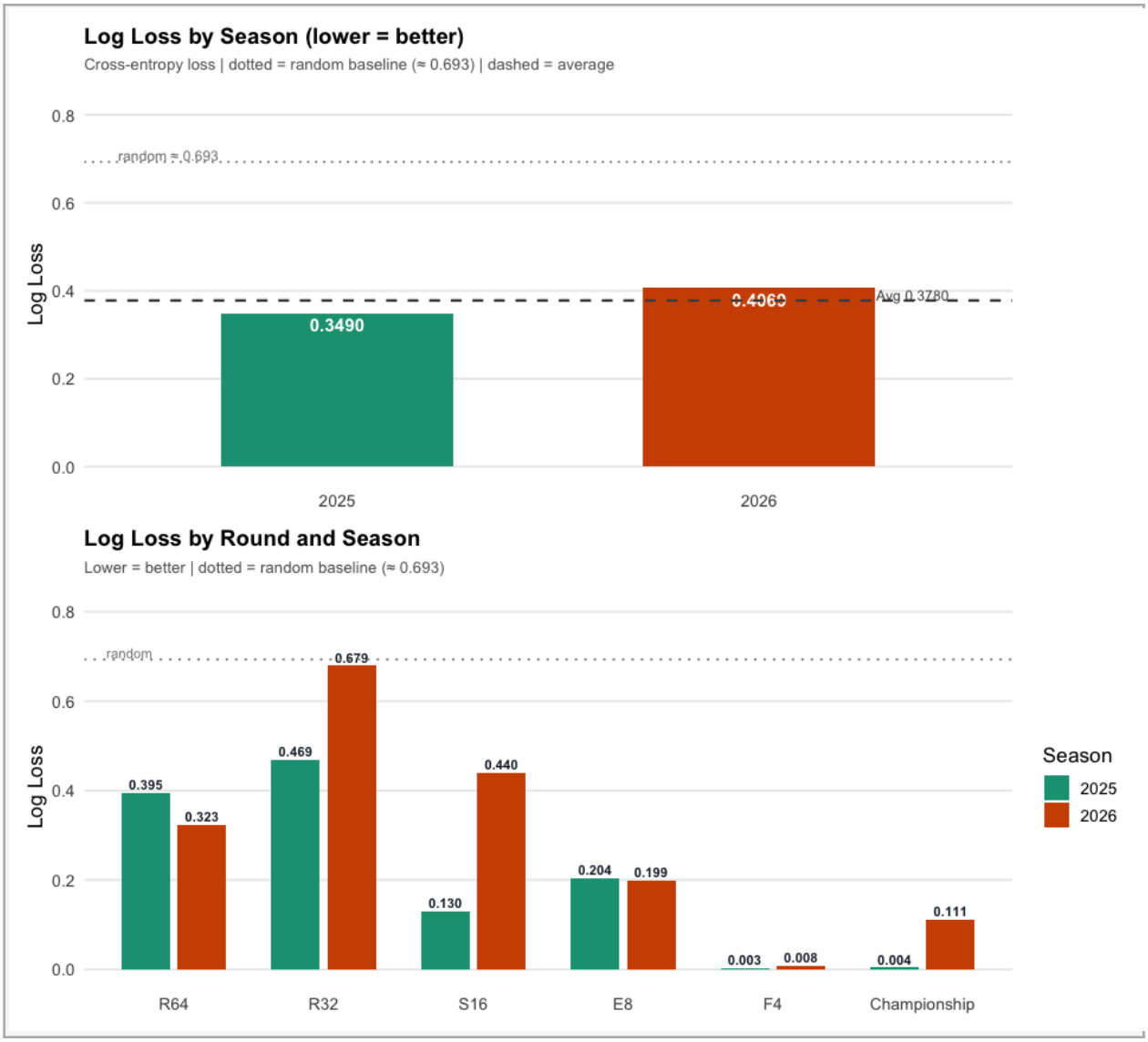


Figure 7. Brier Score by Round (2025 & 2026)

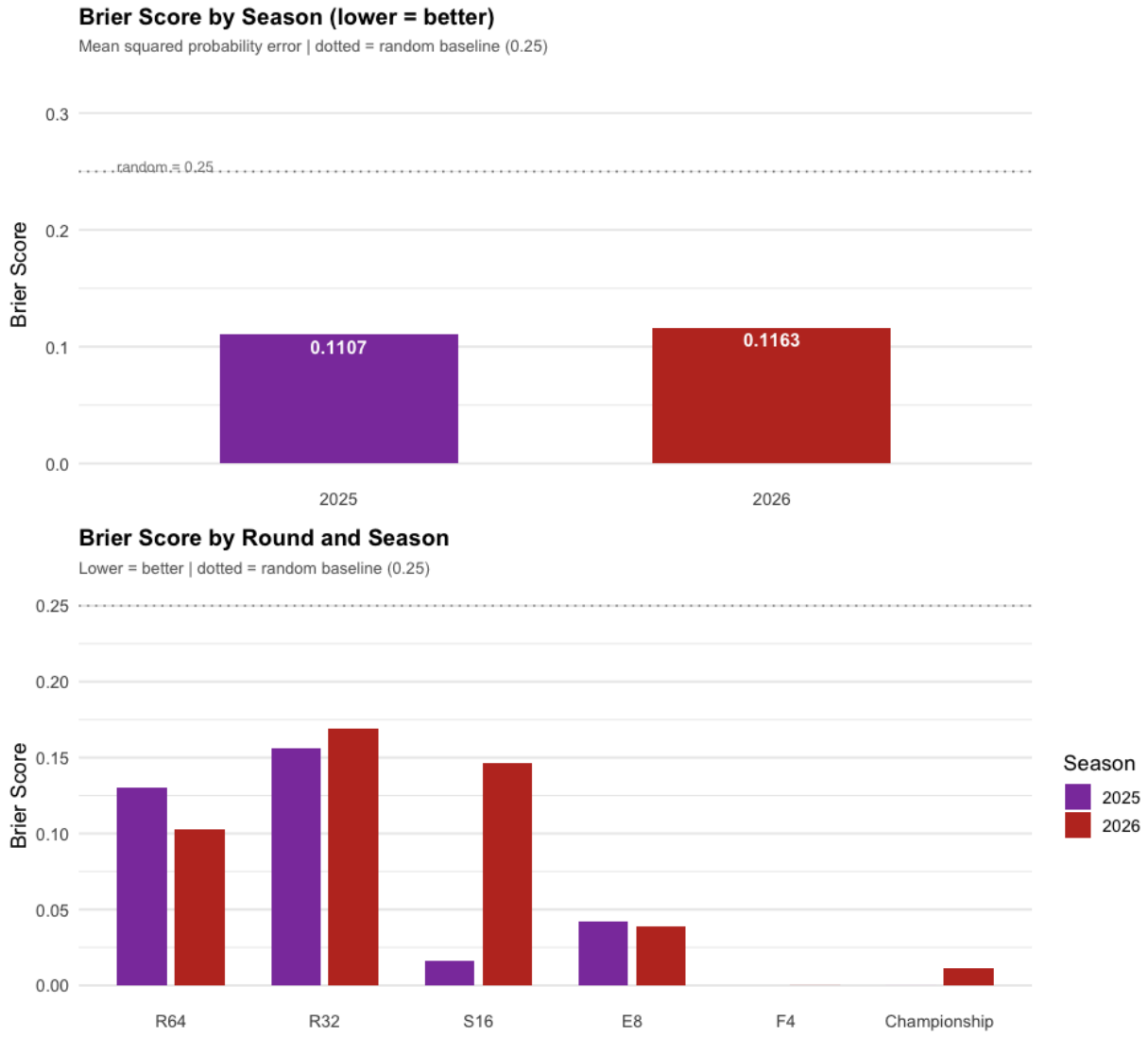


Figure 8. ROC Curves (2025 & 2026)

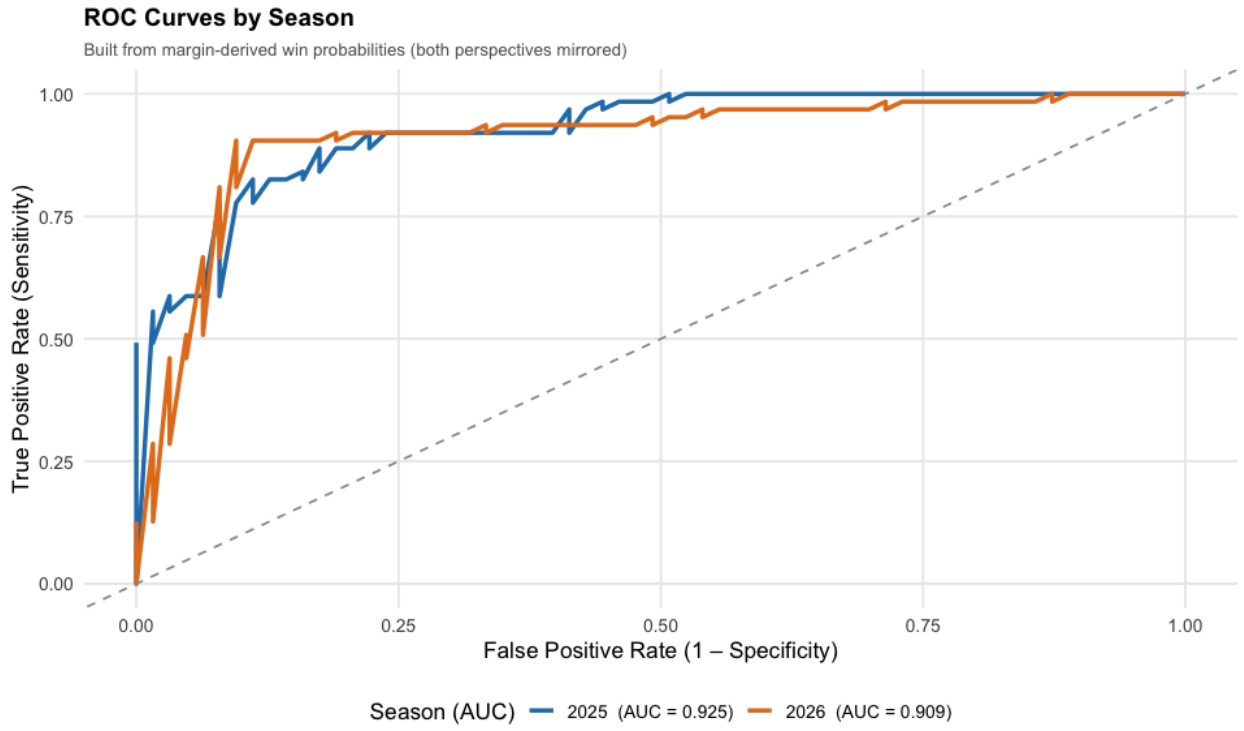


Figure 9. MAE by Round (2025 & 2026)

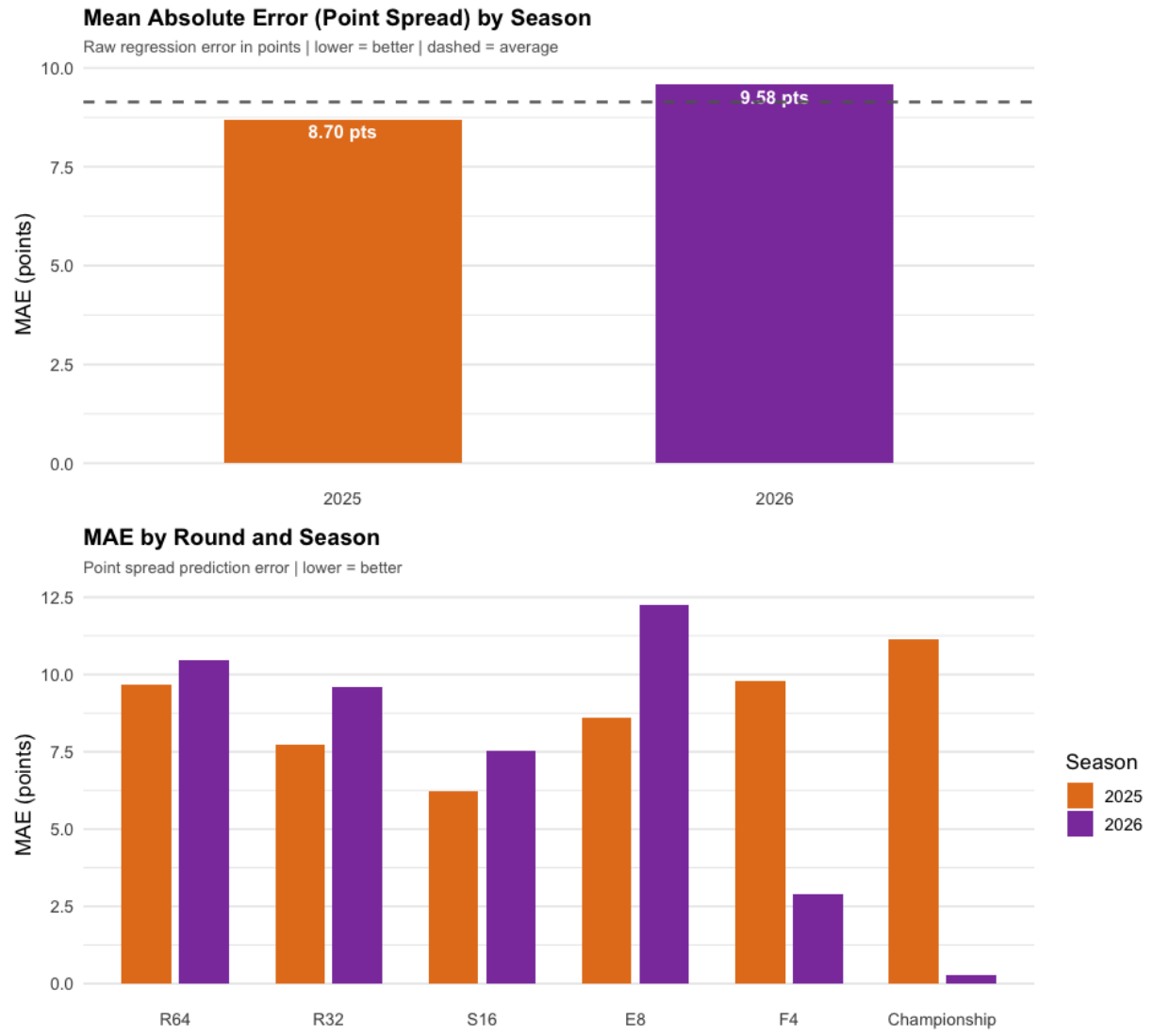


Figure 10. Accuracy by Season (2003-2026)

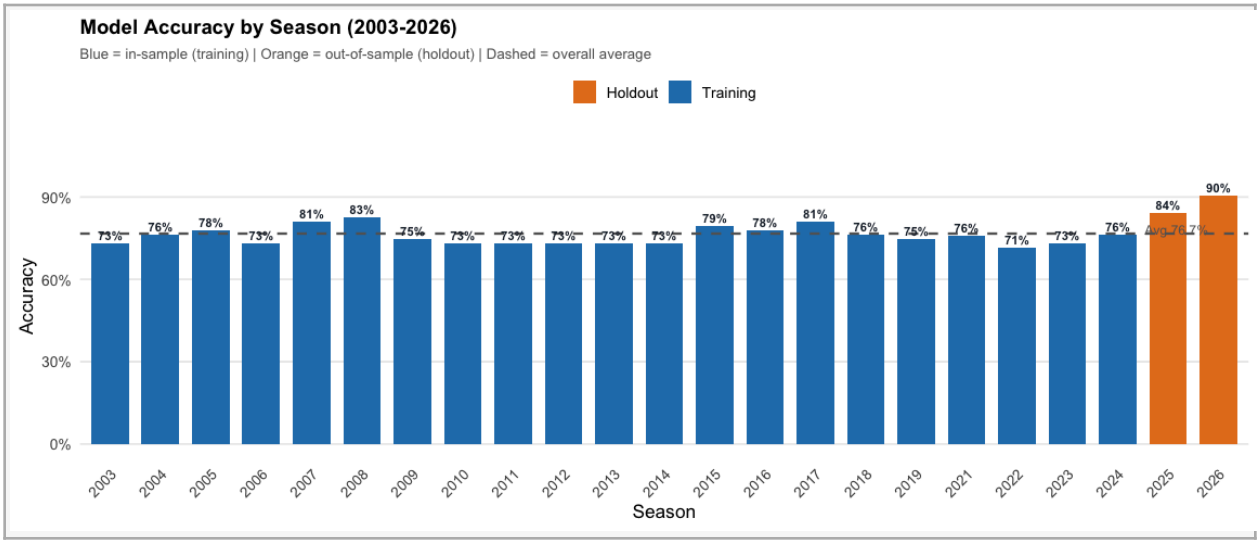


Figure 11. Brier Score by Season (2003-2026)

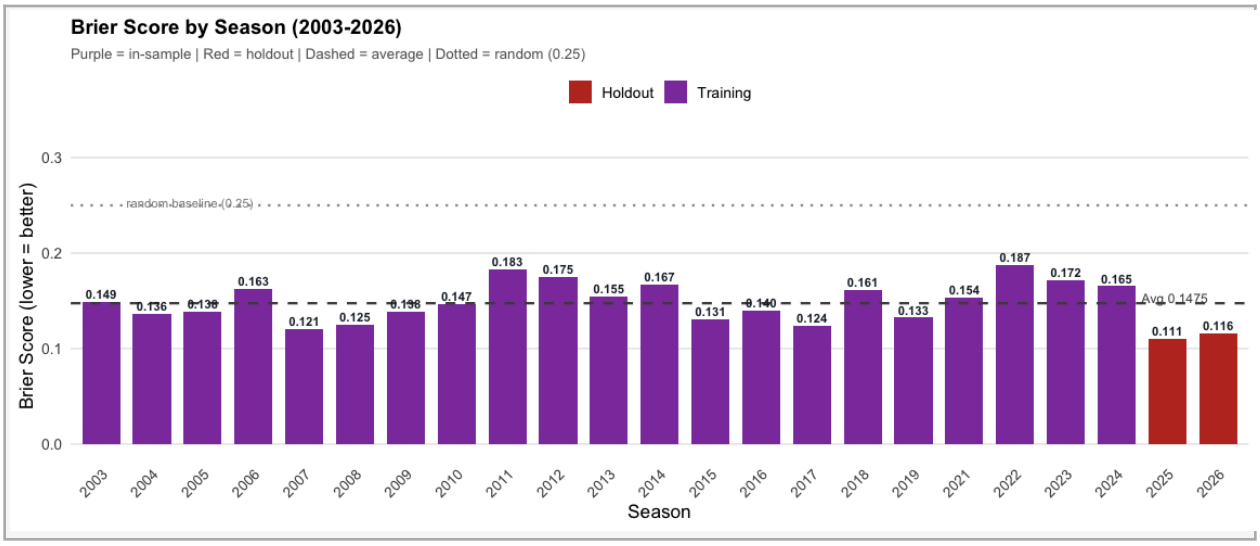


Figure 12. MAE by Season (2003-2026)

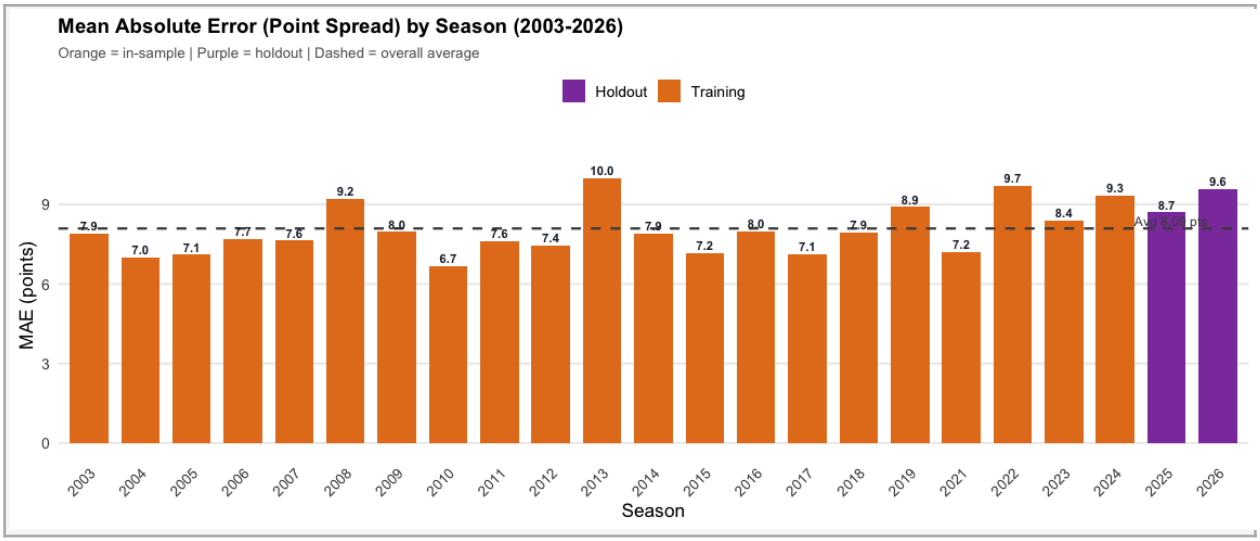


Figure 13. Log-Loss by Season (2003-2026)

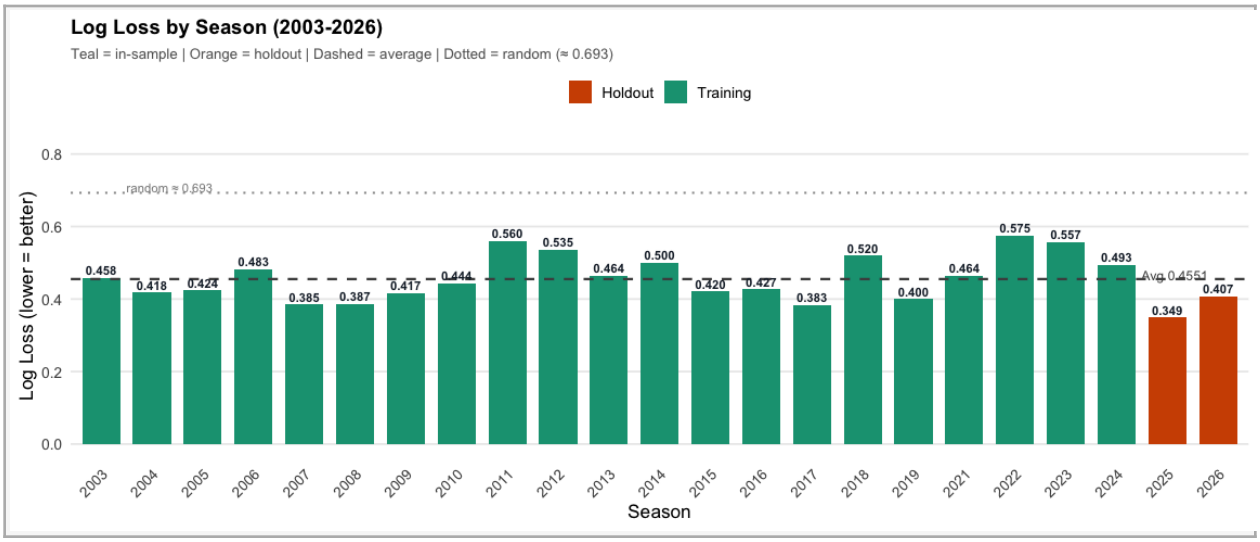


Figure 14. Overall Results by Season (2003-2026)

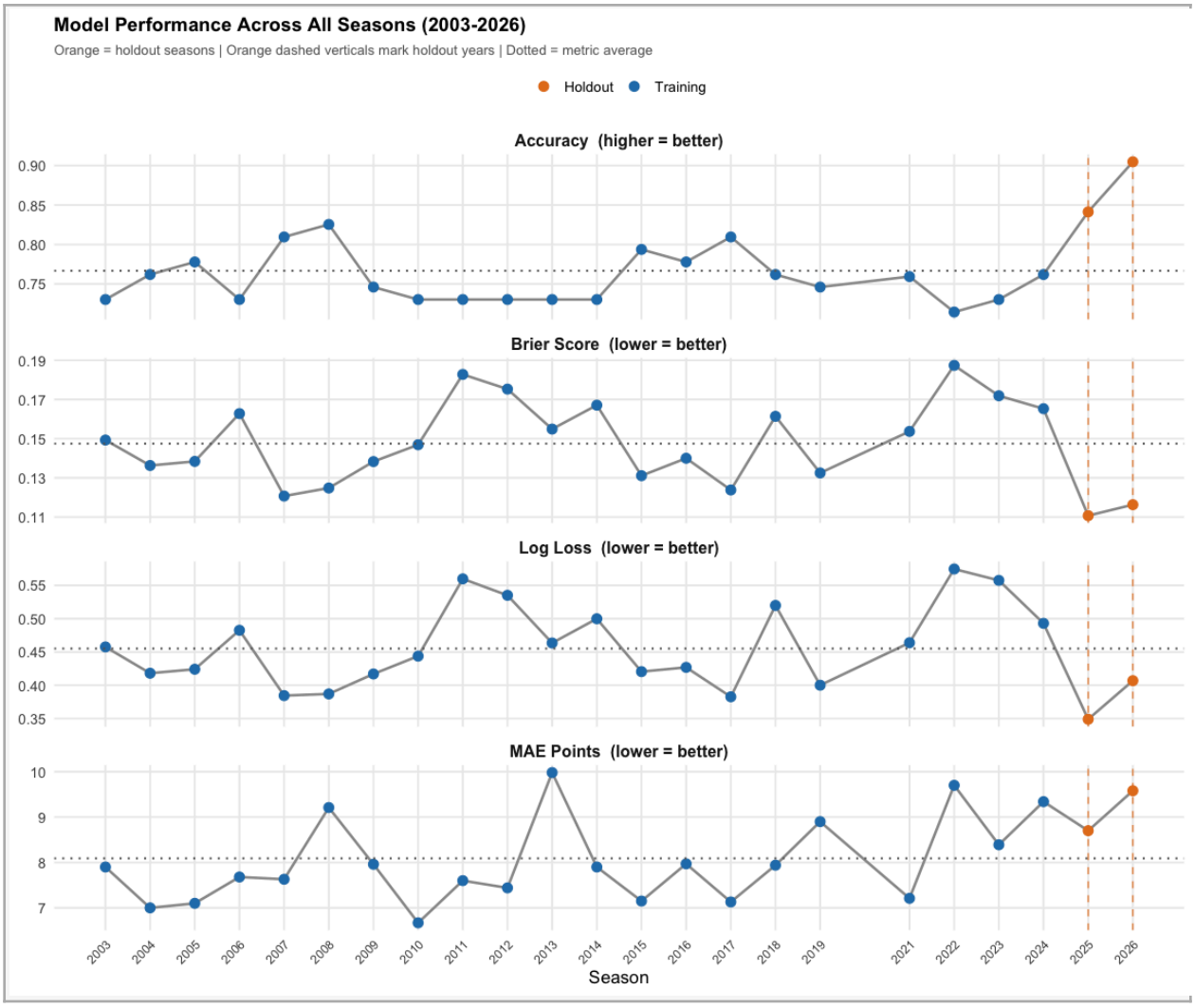


Figure 15. Predicted vs. Actual Margin

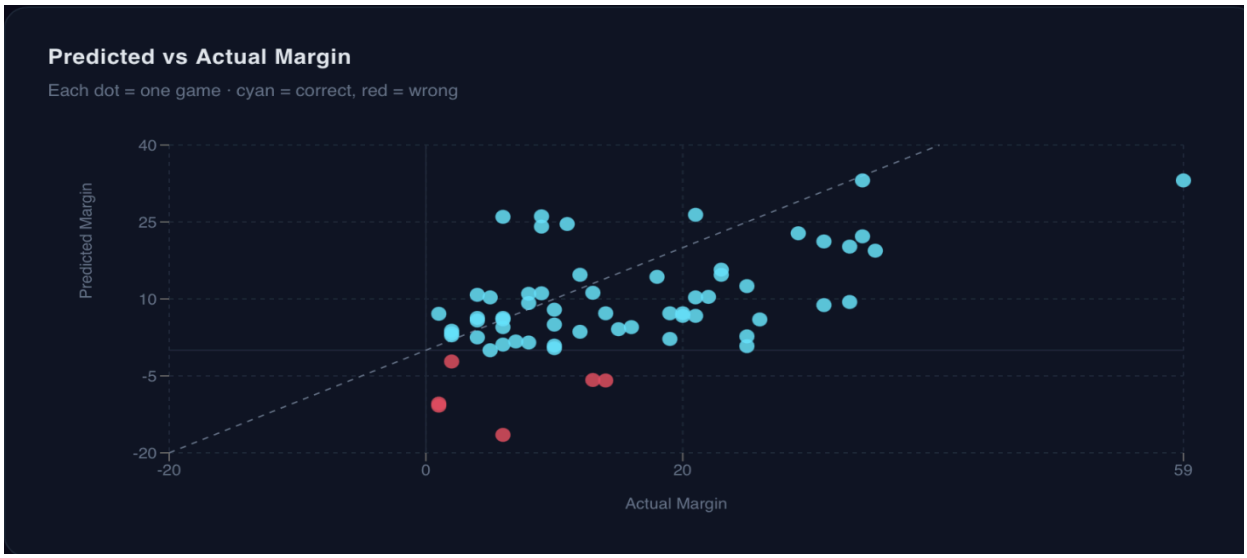
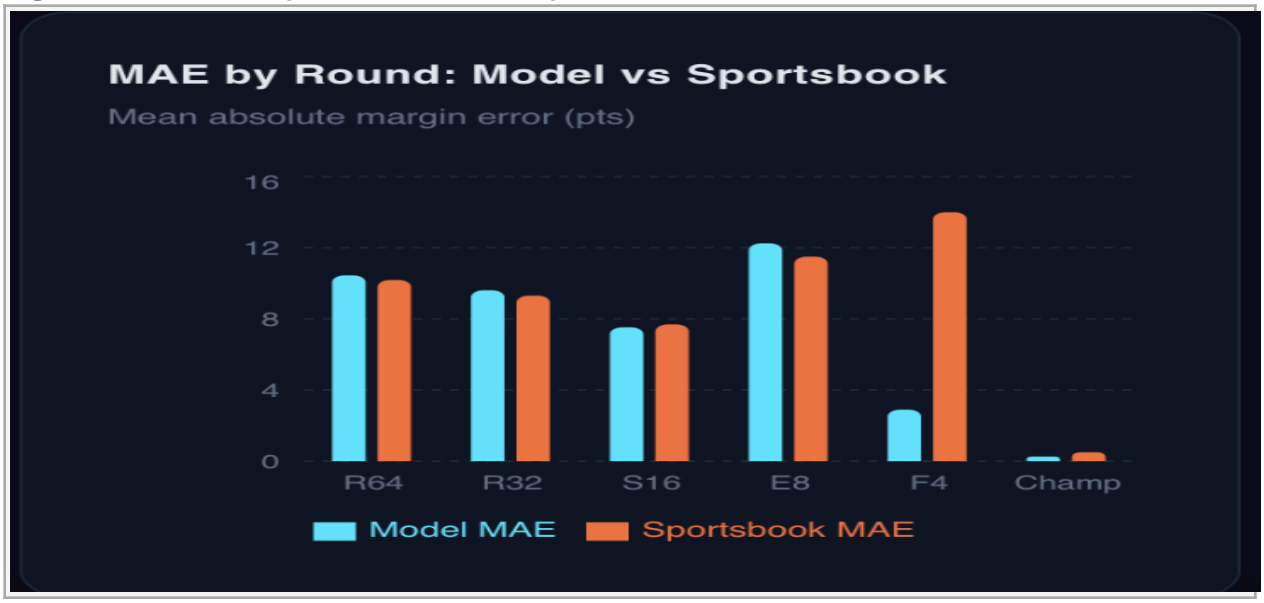


Figure 16. Model vs. Sportsbook MAE Comparison



Dashboards

Dashboard 1. Model vs. Sportsbook Comparison



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